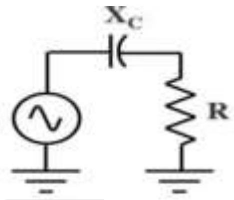
Module-1(b) - BJT AC Models

Syllabus - Base Biased Amplifier, Emitter Biased Amplifier, Small Signal Operation, AC Beta, AC Resistance of the emitter diode, Two transistor models, Analyzing an amplifier, h - parameters, relations b/n r and h parameters. Voltage Amplifiers: Voltage gain, loading effect of input impedance. . CC Amplifier, Output Impedance.

Coupling Capacitors and Bypass Capacitors

Q1. What is the role of Coupling and bypass capacitors in electronic amplifiers? Explain how Coupling Capacitors and Bypass Capacitors are chosen.

Coupling capacitors are used to couple an input ac signal into an amplifier and output of the amplifier to an external load without disturbing Q point.



For a coupling capacitor to work properly, its reactance must be much smaller than the resistance at the lowest frequency of the ac source.

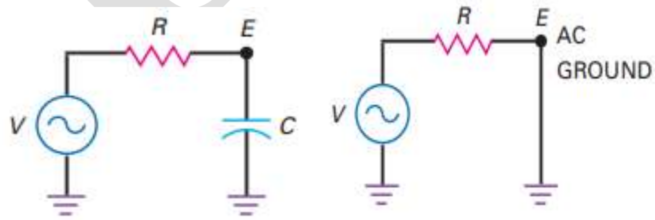
For Good coupling: $\boxed{X_C < 0.1R}$ ------- **10 :1 rule**

The reactance should be at least 10 times smaller than the resistance at the lowest frequency of operation.

Bypass Capacitor

A bypass capacitor it is used to create an ac ground in an electronic amplifier without disturbing its Q point..

When the frequency is high enough, the capacitive reactance is much smaller than the resistance. In this case, almost all the ac source voltage appears across the resistor. Emitter E is effectively shorted to ground.



When used in this way, the capacitor is called a bypass capacitor because it bypasses or shorts point E to ground.

For a bypass capacitor to work properly, its reactance must be much smaller than the resistance at the lowest frequency of the ac source.

- Good bypassing: $\boxed{X_C < 0.1R}$

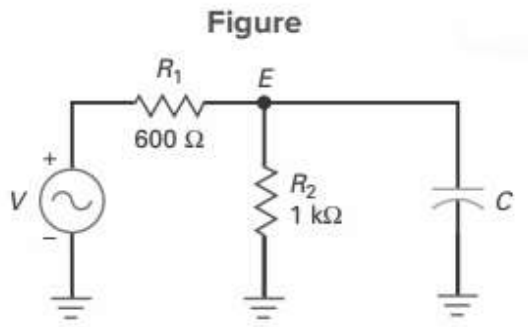
When this rule is satisfied, it can be replaced by short circuit.

Any well-designed circuit satisfies the 10:1 rule, two approximations for a capacitor are

- 1. For dc analysis, the capacitor is open.
- 2. For ac analysis, the capacitor is shorted.

Problem - In Fig shown, the input frequency of V is 1 kHz (i) What value of C is needed to effectively short point E to ground? (ii) Find the value of C needed if R_1 is 50Ω

Solution – (i) For R_1 = 600Ω



Thevenin's resistance seen form the capacitor side is given by

R_{Th} = 600||1000 = 375Ω

Using 10:1 rule for good bypassing - at f = 1kHz

$X_C \le 0.1R_{Th}$ = 37.5 Ω

$$C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi(37.5)(1000)} = 4.2\mu F$$

(ii) For R_1 = 50Ω

Thevenin's resistance seen form the capacitor side is given by

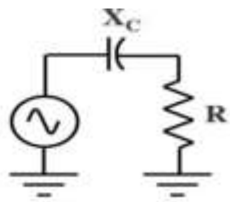
R_{Th} = 50||1000 = 47.61Ω

Using 10:1 rule for good bypassing - at f = 1kHz

$X_C \le 0.1R_{Th}$ = 4.76 Ω

Problem – (i) If R = 2kΩ and the frequency range is from 20 Hz to 20 kHz, find the value of C needed to act as a good coupling capacitor. (ii) Find the value of C when the lowest frequency is 1 kHz and R is 1.6 kΩ.

Solution – (i) Using 10 : 1 rule; At f = 20 Hz, $X_C \le 0.1R = 0.1(2000) = 200\Omega$



$$C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi(20)(200)} = 39.8\ \mu F$$

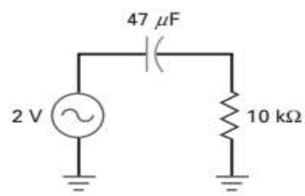
(ii) Using 10 : 1 rule; At f = 1000 Hz, $X_C \le 0.1R = 0.1(1600) = 160\Omega$

$$C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi(1000)(160)} = 1\ \mu F$$

Problem – For the RC circuit shown in figure (i) What is the lowest frequency at which good coupling exists? (ii) If the load resistance is changed to 1 kΩ, what is the lowest frequency for good coupling? (iii) If the capacitor is changed to 100 µF, what is the lowest frequency for good coupling? (iv) If the lowest input frequency is 100 Hz, what C

Solution – (i) Using 10 : 1 rule; At f = f_L, $X_C \le 0.1R = 0.1(10000) = 1000\Omega$

$$f_L = \frac{1}{2\pi C X_C} = \frac{1}{2\pi(47\times10^{-6})(1\times10^{3})} = 3.38\ Hz$$



(ii) Using 10 : 1 rule; At f = f_L, $X_C \le 0.1R = 0.1(1000) = 100\Omega$

$$f_L = \frac{1}{2\pi C X_C} = \frac{1}{2\pi(47\times10^{-6})(100)} = 33.8\ Hz$$

(iii) Using 10 : 1 rule; At f = f_L, $X_C \le 0.1R = 0.1(10000) = 1000\Omega$

$$f_L = \frac{1}{2\pi C X_C} = \frac{1}{2\pi(100\times10^{-6})(1000)} = 1.59\text{ Hz}$$

(iv) Using 10 : 1 rule; At f_L,=100Hz, $X_C \le 0.1R = 0.1(10000) = 1000\Omega$

$$C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi(100)(1000)} = 1.6\mu F$$

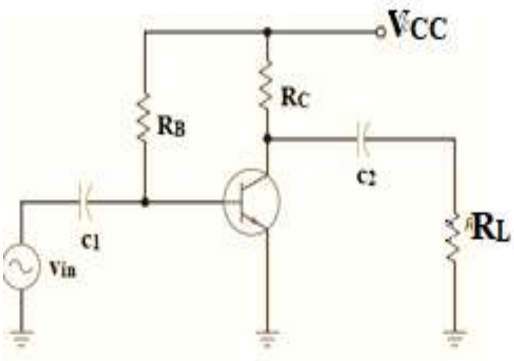
$$C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi(4.76)(1000)} = 33.43\mu F$$

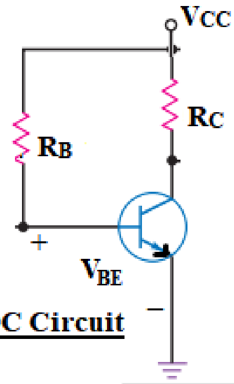
Base Biased Amplifier

Q2. With a neat circuit diagram explain the working of Base biased CE amplifier

A base-biased amplifier has instructional value because its basic ideas can be used to build more complicated amplifiers. Figure shows the circuit diagram of a base biased amplifier and its DC bias circuit.

Base biased amplifer





DC Circuit

DC Analysis

Applying KVL to Base loop,

$V_{CC} - I_B R_B - V_{BE} = 0$

$I_B = \frac{V_{CC} - V_{BE}}{R_B}$ and $I_C = \beta I_B$

KVL to Collector emitter loop,

$V_C = V_{CE} = V_{CC} - I_C R_C$

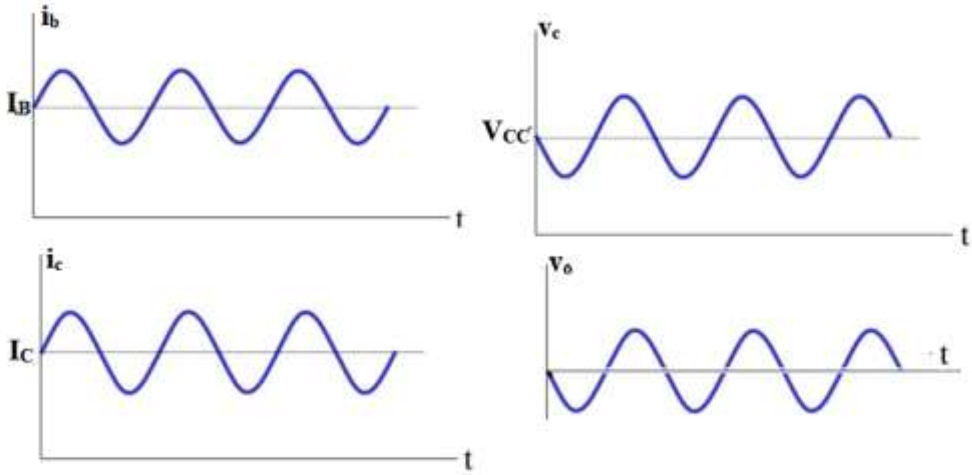
The Q point is located at (V_{CEQ}, I_{CQ})

Circuit Operation

The coupling capacitors C_1 and C_2 couples the input to the amplifier and output of the amplifier to the load respectively and prevent the ac source and load resistance from changing the Q point.

The input ac voltage produces an ac base current that is added to the existing dc base current. In other words, the total base current will have a dc component and an ac component. An ac component is superimposed on the dc component. On the positive half-cycle, the ac base current adds to the dc base current I_B and on the negative half-cycle, it subtracts from it. The ac base current produces an amplified variation in collector current because of the current gain.

The amplified collector current flows through the collector resistor, produces a varying voltage across the collector resistor R_C. The collector voltage is swinging sinusoidally above and below the dc level of V_C. Also, the ac collector voltage is inverted (180° out of phase with the input voltage). The output coupling capacitor couples the ac collector voltage to the load resistor. The load voltage is a pure ac signal with an average value of zero. The current and voltage waveforms are shown below.



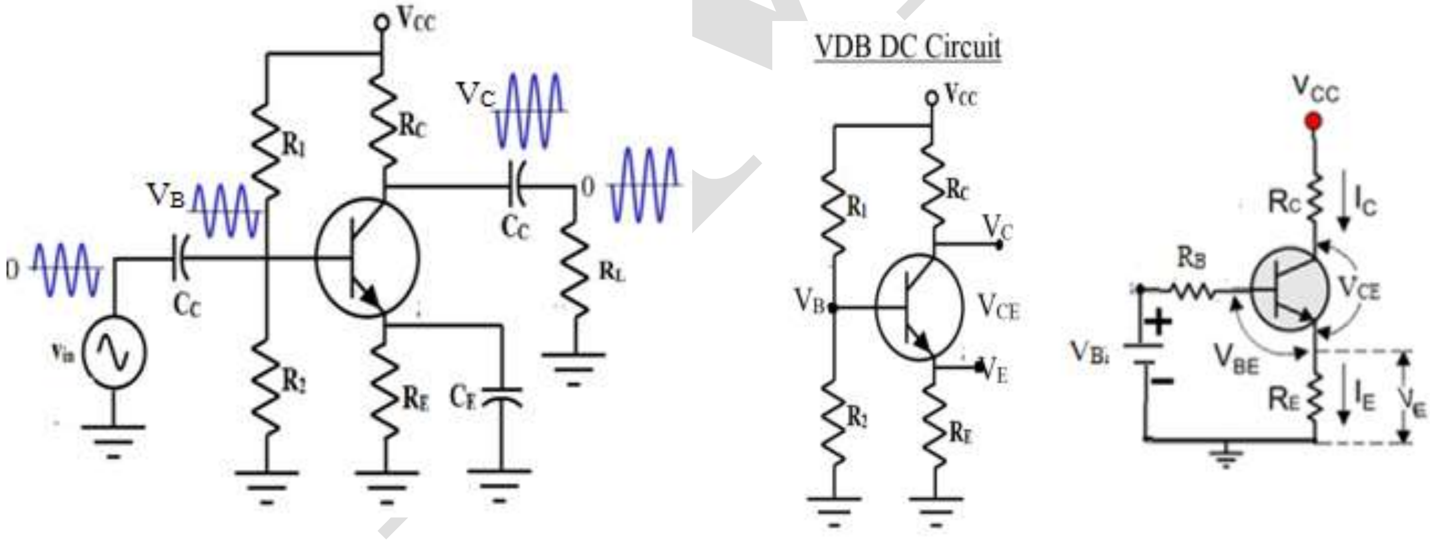
Voltage Gain

The voltage gain of an amplifier is defined as the ratio of ac output voltage to the ac input voltage. $A_V = \frac{v_0}{v_{in}}$

Emitter Biased Amplifier or Voltage Divider Bias (VDB) Amplifier

Q3. With a neat circuit diagram explain the working of Base biased CE amplifier

An emitter-biased amplifier which gives stable Q point is widely used instead of base biased amplifier. Figure shows a VDB amplifier and its corresponding DC circuit.



DC Analysis

$R_B = R_{Th} = R_1 || R_2$ $\qquad V_{BB} = V_{Th} = \frac{V_{CC} R_2}{R_1 + R_2}$

Applying KVL to base-emitter loop $\quad V_B - I_B R_B - V_{BE} - I_E R_E = 0$

Using $I_B = \frac{I_E}{(\beta+1)}$ in the above equation results

$I_E = \dfrac{V_{BB} - V_{BE}}{R_E + \dfrac{R_B}{\beta + 1}}$	$I_B = \dfrac{I_E}{\beta + 1} = \dfrac{V_{BB} - V_{BE}}{R_B + R_E(\beta + 1)}$	$I_C = \beta\, I_B$

Applying KVL to Collector-emitter loop $V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$

Using $I_E \cong I_C$ in the above equation,

$V_{CE} = V_{CC} - I_C(R_C + R_E)$ or $I_C = \frac{V_{CC} - V_{CE}}{R_C + R_E}$

The Q point is located at (V_{CEQ}, I_{CQ})

General thumb rule is that the voltages are selected as $V_{CE} = 0.5V_{CC}$; $V_E = 0.1V_{CC}$ for most of the requirements

.**Circuit Operation**

Coupling capacitors are used between the source and base as well as between the collector and the load resistance. The bypass capacitor is used between the emitter and ground facilitates larger voltage gain by coupling the entire ac input voltage v_{in} into the base and all of this ac voltage appears across the base-emitter diode. The ac base current then produces an amplified ac collector voltage.

The ac input voltage is a small sinusoidal voltage with an average value of zero. The base voltage is an ac voltage superimposed on a dc voltage of V_B.

The collector voltage is an amplified and inverted ac voltage superimposed on the dc collector voltage of V_C. The load voltage is the same as the collector voltage, except that it has an average value of zero.

Voltage Gain

The voltage gain of an amplifier is defined as the ratio of ac output voltage to the ac input voltage. $A_V = \frac{v_O}{v_{in}}$

Q4. Define (a) Small-Signal current (b) DC current gain (c) AC current gain (d) AC resistance of the emitter diode

The total emitter current consists of a dc component and an ac component.

$I_E = I_{EQ} + i_e$

Due to higher values of i_e, the Q point can move to either saturation or cut-off which results in output distortion (clipping of peaks). To minimize distortion, the peak-to-peak value of i_e must be small compared to I_{EQ}.

- For small-signal operation
- $i_{e(p-p)} < 0.1 I_{EQ}$ ------ **10 percent rule**

The Current Gain

The Current gain or DC current gain is defined as

$$\beta_{dc} = \frac{\text{Collector Current at Q point}}{\text{Corresponding Base current}} = \frac{I_C}{I_B}$$

The ac current gain is different and is defined as

$$\beta_{ac} = \beta = \frac{\text{ac collector current } i_C}{\text{ac base current } i_b}$$

the value of the ac current gain is different from the dc current gain. ($\beta_{dc} \neq \beta_{ac}$)

AC Resistance of the Emitter Diode($r^{|}_e$)

When a small ac voltage is across the emitter diode, it produces the ac emitter current. The value of this ac emitter current depends on the location of the Q point.

The total emitter current has a dc component and an ac component and similarly Base to emitter voltage.

$$I_E = I_{EQ} + i_e \qquad\qquad V_{BE} = V_{BEQ} + v_{be}$$

The sinusoidal variation in V_{BE} produces a sinusoidal variation in I_E.

The ac emitter resistance of the emitter diode $r_e^{|}$ equal to the ac base-emitter voltage v_{be} divided by the ac emitter current i_e.

$$r_e^{|} = \frac{v_{be}}{i_e} \text{ ohms}$$

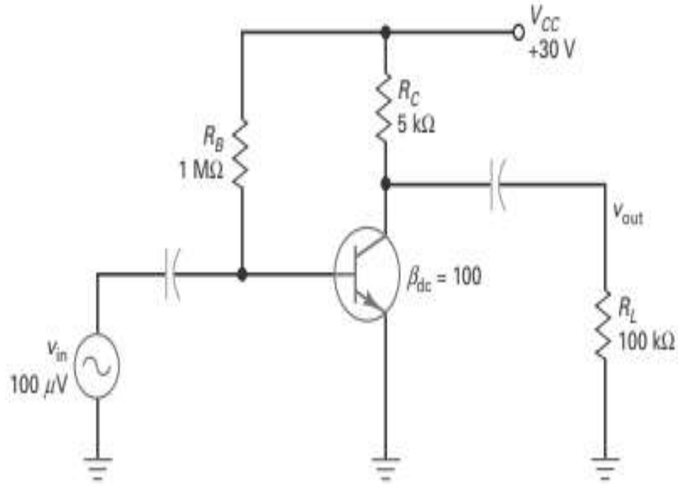
Note - The ac emitter resistance always decreases when the dc emitter current increases because v_{be} is essentially a constant value.

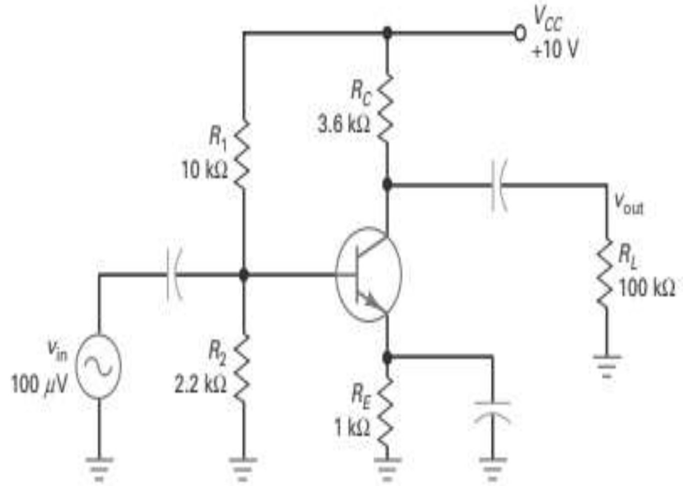
Using solid-state physics and calculus, the ac emitter resistance is approximated as

$$r_e^{|} = \frac{25 \text{ mV}}{I_E} \quad \text{ohms}$$

The reason r_e' is important is that it determines the voltage gain. The smaller it is, the higher the voltage gain.

Problem – For the circuits shown, calculate the (i) maximum small signal current (ii) emitter resistance.





Solution

For the Base bias circuit	For the VDB circuit
$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{30 - 0.7}{1 \times 10^6} = 29.3\mu A$ Assuming β=100 and V_{BE} = 0.7V	$R_B = R_{Th} = R_1 \| \| R_2 = 1.8\ k\Omega$ $V_B = V_{Th} = \frac{V_{CC} R_2}{R_1 + R_2} = 1.8$ V Assuming β=100 and V_{BE} = 0.7V

$I_C \cong I_E \cong \beta I_B = 100(29.3\mu A) = 2.9\ mA$ Using the 10 percent rule for small-signal operation, small signal current $i_{e(p-p)} < 0.1 I_{EQ}$ $i_{e(p-p)} \le 0.1(2.9) = 290\ \mu A$ Emitter resistance $r_e^{\mid} = \frac{25\ mV}{I_E}$ $r_e^{\mid} = \frac{25\ mV}{2.9 \times 10^{-3}} = 8.6\Omega$	$I_E = \frac{V_B - V_{BE}}{R_E + \frac{R_B}{\beta + 1}} = \frac{1.8 - 0.7}{1 + \frac{1.8}{101}} = 1.08\ mA$ Using the 10 percent rule for small-signal operation, small signal current $i_{e(p-p)} < 0.1 I_{EQ}$ ➔ $i_{e(p-p)} \le 0.1(1.08) = 108\ \mu A$ Emitter resistance $r_e^{\mid} = \frac{25\ mV}{I_E}$ $r_e^{\mid} = \frac{25\ mV}{1.08 \times 10^{-3}} = 23.15\Omega$

<u>Two Transistor Models</u>

Q5. Explain the transistor models

To analyze the ac operation of a transistor amplifier, an ac-equivalent circuit or a model for the transistor that simulates how it behaves when an ac signal is present is required.

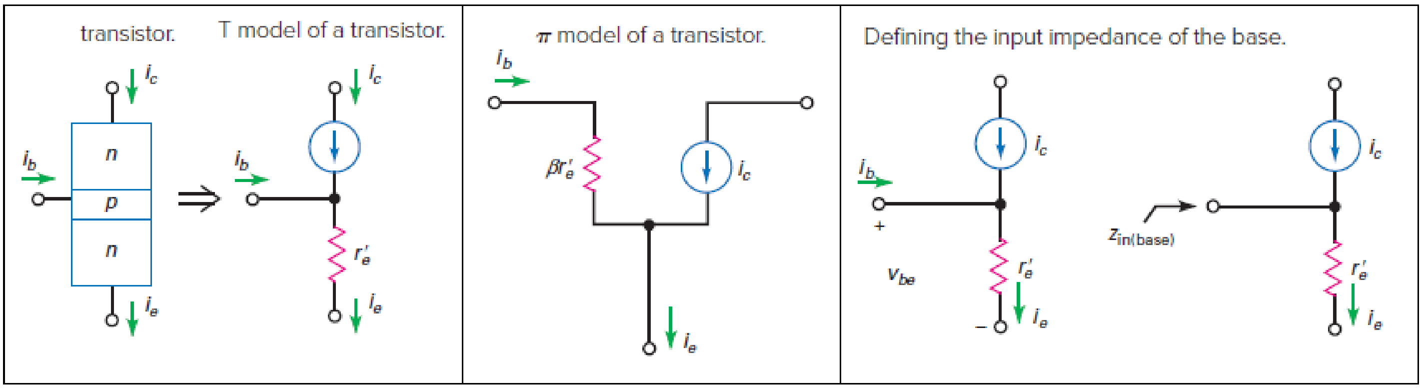
There are two models are primarily used

(i) The T Model (ii) The π Model

<u>The T Model (</u>Ebers-Moll model)

As far as a small ac signal is concerned, the emitter diode of a transistor acts like an ac resistance r'_e and the collector diode acts like a current source i_c and it is represented as T model.

When an ac input signal drives a transistor amplifier, an ac base-emitter voltage v_{be} is



across the emitter diode, this produces an ac base current i_b.

The ac voltage source has to supply this ac base current so that the transistor amplifier will work properly. Stated another way, the ac voltage source is loaded by the input impedance of the base. Looking into the base of the transistor, the ac voltage source sees an input impedance $z_{in(base)}$.

Applying Ohm's law to the emitter diode $v_{be} = i_e\, r_e^{|}$

At low frequencies, this impedance is purely resistive and defined as

$$z_{in\,(base)} = \frac{v_{be}}{i_b} = \frac{i_e\, r_e^{|}}{i_b} = \beta r_e^{|} \qquad \text{--------(A)}$$

<u>The π Model</u>

The equation (A) is implemented using the π model. The π model is easier to use than the T model because the input impedance is not obvious when you look at the T model.

<u>Analyzing an Amplifier</u>

<u>Q6. Explain the procedure for analyzing the amplifiers</u>

Amplifier analysis is complicated because both dc and ac sources are in the same circuit. To analyze amplifiers, the effect of the dc sources and ac sources are to be considered using the principle of superposition. The analysis split into two parts: a dc analysis and an ac analysis.

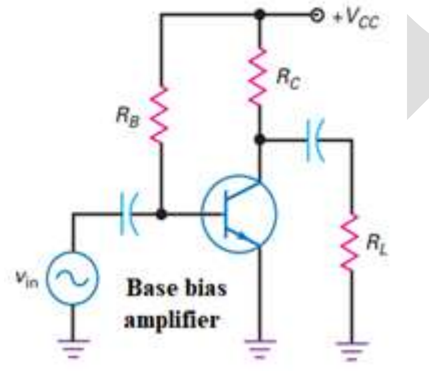
<u>In the dc analysis</u>, the dc voltages and currents are calculated considering all capacitors are open circuit.. Using the value of dc emitter current, the emitter resistance r'$_e$ required for the ac analysis is determined.

<u>In ac analysis</u>, (a) Short all coupling and bypass capacitors. (b) Visualize all dc supply voltages as ac grounds. (c) Replace the transistor by its T model or π Model. (d) Draw the ac-equivalent circuit and determine the voltage gain of the amplifier.

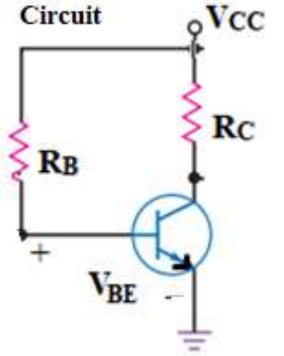
<u>Base Biased Amplifier</u>

Q7. Explain the analysis of Base biased amplifier using (i) π– model (ii) T – Model and derive the expression for voltage gain

Figure shows the Base biased amplifier, its DC equivalent and ac equivalent circuit for analysis.



DC Equivalent Circuit



<u>DC Analysis</u>

$I_B = \frac{V_{CC} - V_{BE}}{R_B}$ $\qquad I_C \cong I_E \cong \beta I_B$

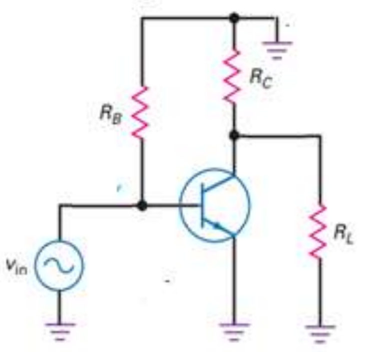
$V_{CE} = V_{CC} - I_C R_C$ $\qquad$ Q Point (V_{CEQ}, I_{CQ})

Using the 10 percent rule,

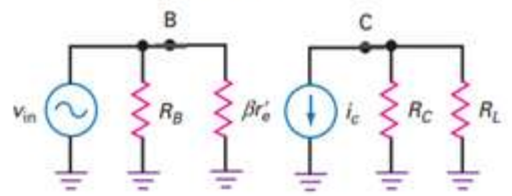
Maximum small signal current $i_{e(p\text{-}p)} < 0.1 I_{EQ}$

Emitter resistance $r_e^{|} = \frac{25\text{ mV}}{I_E}$

ac equivalent circuit



π Model



<u>**AC Analysis**</u>

Applying Ohm's law to the base circuit,

$v_{in} = i_b\ \beta r_e^{|}$ --------(1)

The collector acts a current source of i_C and this current flows through the ac collector resistance $r_c = (R_C||R_L)$

The output voltage is given by

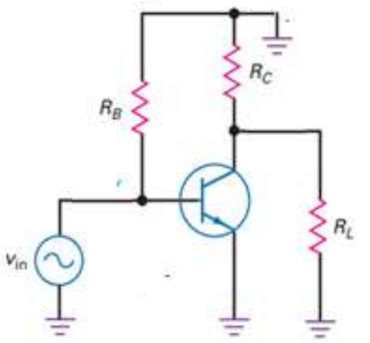
$V_{out} = i_c(R_C||R_L) = \beta i_b\, r_C$ -------(2)

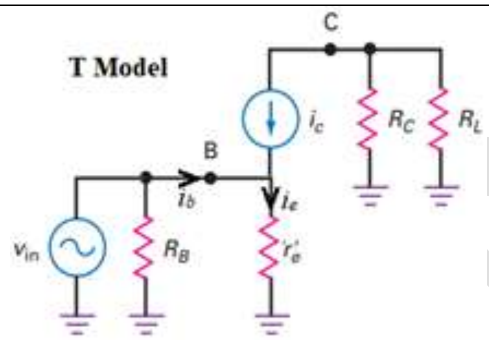
From equations (1) and (2)

$$\text{Voltage gain } A_V = \frac{v_{out}}{v_{in}} = \frac{\beta i_b r_c}{i_b \beta r_e^{|}}$$

$$A_V = \frac{r_c}{r_e^{|}}$$

ac equivalent circuit





<u>**AC Analysis**</u>

By Ohm's law to the base circuit,

$v_{in} = i_e r_e^{|} \cong \beta\ i_b r_e^{|}$ --------(1)

The collector acts a current source of i_C and this current flows through the ac collector resistance $r_c = (R_C||R_L)$

$V_{out} = i_c(R_C||R_L) = \beta i_b\, r_C)$

$$\text{Voltage gain } A_V = \frac{v_{out}}{v_{in}} = \frac{\beta i_b r_c}{i_b \beta r_e^{|}}$$

$$A_V = \frac{r_c}{r_e^{|}}$$

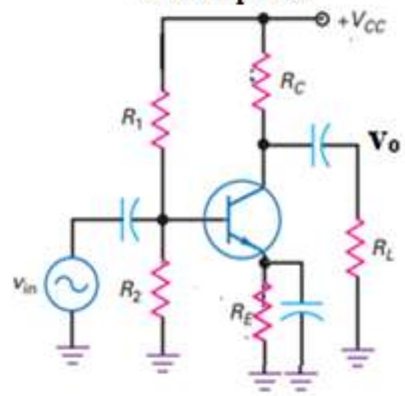
<u>Voltage Divider Bias (VDB) Amplifier</u>

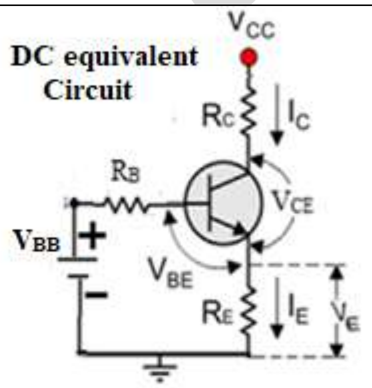
Q8. Explain the analysis of VDB amplifier using (i) π– model (ii) T – Model and derive the expression for voltage gain.

Figure shows the VDB amplifier, its DC equivalent and ac equivalent circuit for analysis.

<u>**VDB Amplifier -**</u>Figure shows the ac equivalent circuits for a VDB amplifier.

VDB anplifier





<u>**DC Analysis**</u>

$R_B = R_{Th} = R_1 || R_2$

and $V_{BB} = V_{Th} = \frac{V_{CC} R_2}{R_1 + R_2}$

<u>By Ohm's law to base circuit,</u>

$V_{BB} - I_B R_B - V_{BE} - I_E R_E = 0$

$I_E \cong \frac{V_B - V_{BE}}{R_B + \beta R_B}$ [using $I_E \cong \beta\ I_B$]

$I_C = \beta I_B$ and $I_E = I_B + I_C$

<u>KVL to Collector emitter loop</u>

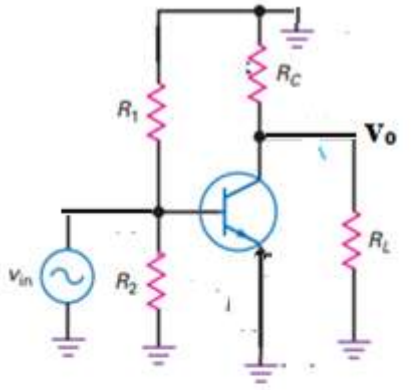
$V_{CE} = V_{CC} - I_C(R_C + R_E)$

Q Point (V_{CEQ}, I_{CQ})

Using the 10 percent rule,

Maximum small signal current $i_{e(p-p)} < 0.1 I_{EQ}$

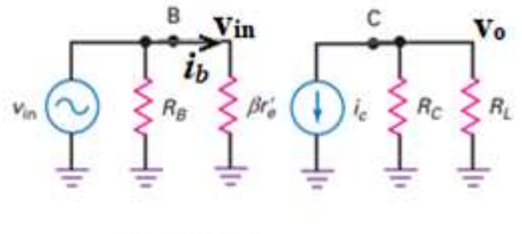
Emitter resistance $r_e^{|} = \frac{25\ mV}{I_E}$



$R_B = R_1 || R_2$

AC Analysis

π Model



$R_B = R_1 || R_2$

Applying Ohm's law to the base circuit,

$v_{in} = i_b\, \beta r_e^{|}$ --------(1)

The collector acts a current source of i_C and this current flows through the ac collector resistance $r_c = (R_C||R_L)$

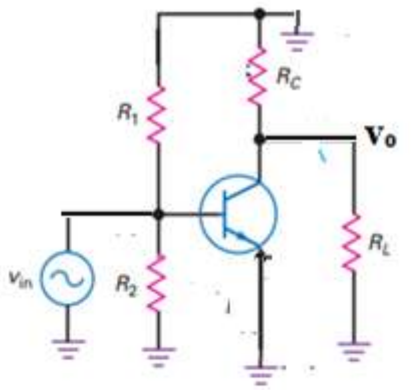
The output voltage is given by

$V_{out} = i_c(R_C||R_L) = \beta i_b\, r_C$ --------(2)

From equations (1) and (2)

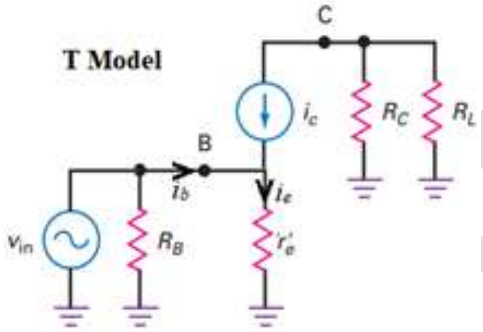
$$\text{Voltage gain } A_V = \frac{v_{out}}{v_{in}} = \frac{\beta i_b r_c}{i_b \beta r_e^{|}}$$

$$A_V = \frac{r_c}{r_e^{|}}$$



$R_B = R_1 || R_2$

T Model



AC Analysis

By Ohm's law to the base circuit,

$v_{in} = i_e r_e^{|} \cong \beta\, i_b r_e^{|}$ --------(1)

The collector acts a current source of i_C and this current flows through the ac collector resistance $r_c = (R_C||R_L)$

$V_{out} = i_c(R_C||R_L) = \beta i_b\, r_C)$

$$\text{Voltage gain } A_V = \frac{v_{out}}{v_{in}} = \frac{\beta i_b r_c}{i_b \beta r_e^{|}}$$

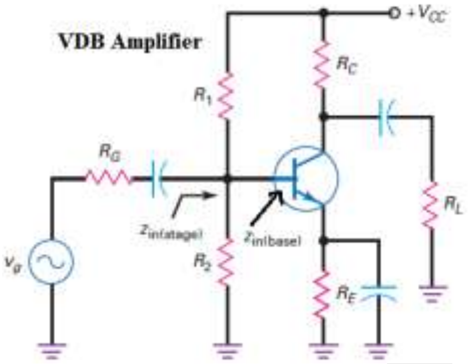
$$A_V = \frac{r_c}{r_e^{|}}$$

The Loading Effect of Input Impedance

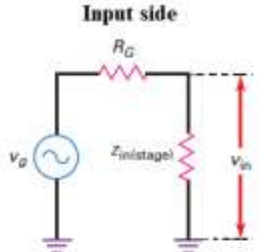
Q9. Explain the effect of loading input impedance in CE amplifiers

Consider the input ac voltage source v_g with an internal resistance of R_G. Some of the ac

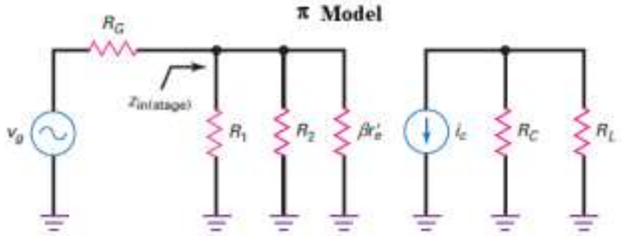
VDB Amplifier



Input side



π Model



source voltage is dropped across its internal resistance R_G. As a result, the ac voltage between the base and ground is less as the ac generator has to drive the input impedance of the stage $z_{in(stage)}$.This input impedance includes the effects of the biasing resistors R_1 and R_2, in parallel with the input impedance of the base $z_{in(base)}$.

$$z_{in\,(stage\,)} = \beta r_e^{|} || (R_1 || R_2)$$

Using voltage divider rule, the input voltage can be expressed as

$$v_{in} = \left[\frac{z_{in\,(stage)}}{R_G + z_{in\,(stage)}}\right] v_g$$

Note – normal value of $R_G < 0.01 z_{in(stage)}$

Problem – (i) Determine the output voltage for the amplifier circuit shown below given that the input source has an internal resistance of 600Ω and β = 300 (ii) Change the R_G to 50Ω and solve for the new amplified output voltage (iii) repeat the calculations for β = 50 and R_G = 600Ω

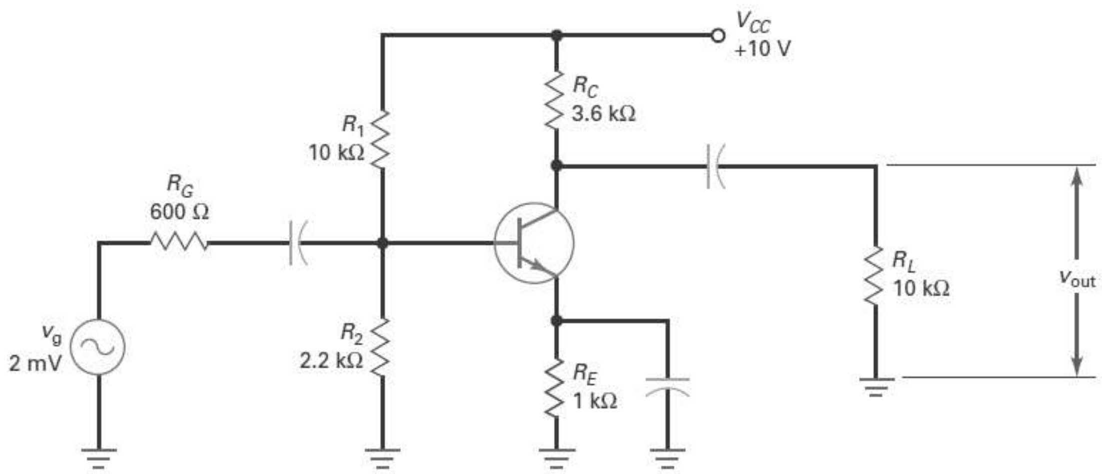
Solution –

$R_B = R_{Th} = R_1 || R_2 = 10 || 2.2 = 1.8$ kΩ

$$V_B = V_{Th} = \frac{V_{CC} R_2}{R_1 + R_2} = \frac{10(2.2)}{10+2.2} = 1.8 \text{ V}$$

Assuming V_{BE} = 0.7V and given β=300

$$I_E = \frac{V_B - V_{BE}}{R_E + \frac{R_B}{\beta+1}} = \frac{1.8-0.7}{1+\frac{1.8}{301}} = 1.093 \text{ mA}$$



Using the 10 percent rule for small-signal operation

Small signal current $i_{e(p-p)} < 0.1 I_{EQ}$ ➔ $i_{e(p-p)} \leq 0.1(1.093) = 109.3$ µA

$$r_e^{|} = \frac{25 \text{ mV}}{I_E} = \frac{25 \text{ mV}}{1.093 \times 10^{-3}} = 22.86\ \Omega$$

ac collector resistance $r_c = (R_C || R_L) = (3.6||10) = 2.64\text{k}\Omega$

$$\text{Voltage gain } A_V = \frac{r_c}{r_e^{|}} = \frac{2640}{22.86} = 116$$

$$z_{in\,(stage)} = \beta r_e^{|} || (R_1 || R_2) = 300(22.86)||[1.8] = 1.42\text{k}\Omega$$

<u>Given R_G = 600Ω</u>

$$v_{in} = \left[\frac{z_{in\,(stage)}}{R_G + z_{in\,(stage)}}\right] v_g = \left[\frac{1.42 \times 2 \times 10^{-3}}{0.6 + 1.42}\right] = 1.405 \text{ mV}$$

$$\text{Voltage gain } A_V = \frac{v_{out}}{v_{in}} \quad \text{or} \quad v_{out} = A_V v_{in} = 116(1.405) = 163 \text{ mV}$$

<u>(ii) Given R_G = 50Ω</u>

$$z_{in\,(stage)} = \beta r_e^{|}||(R_1||R_2) = 300(22.86)||[1.8] = 1.42k\Omega$$

$$v_{in} = \left[\frac{z_{in\,(stage)}}{R_G + z_{in\,(stage)}}\right] v_g = \left[\frac{1.42 \times 2 \times 10^{-3}}{0.05 + 1.42}\right] = 1.993\ mV$$

$$\text{Voltage gain } A_V = \frac{v_{out}}{v_{in}} \quad \text{or} \quad v_{out} = A_V v_{in} = 116(1.993) = 231\ mV$$

When β decreases, the input impedance of the base decreases, the input impedance of the stage decreases, the input voltage decreases, and the output voltage decreases.

<u>(iii) Given β = 50 and R_G = 600Ω</u>

$$z_{in\,(stage)} = \beta r_e^{|}||(R_1||R_2) = 50(22.86)||[1.8] = 699\ \Omega$$

$$\text{Given } R_G = 600\Omega\ v_{in} = \left[\frac{z_{in\,(stage)}}{R_G + z_{in\,(stage)}}\right] v_g = \left[\frac{0.699 \times 2 \times 10^{-3}}{0.6 + 0.699}\right] = 1.07\ mV$$

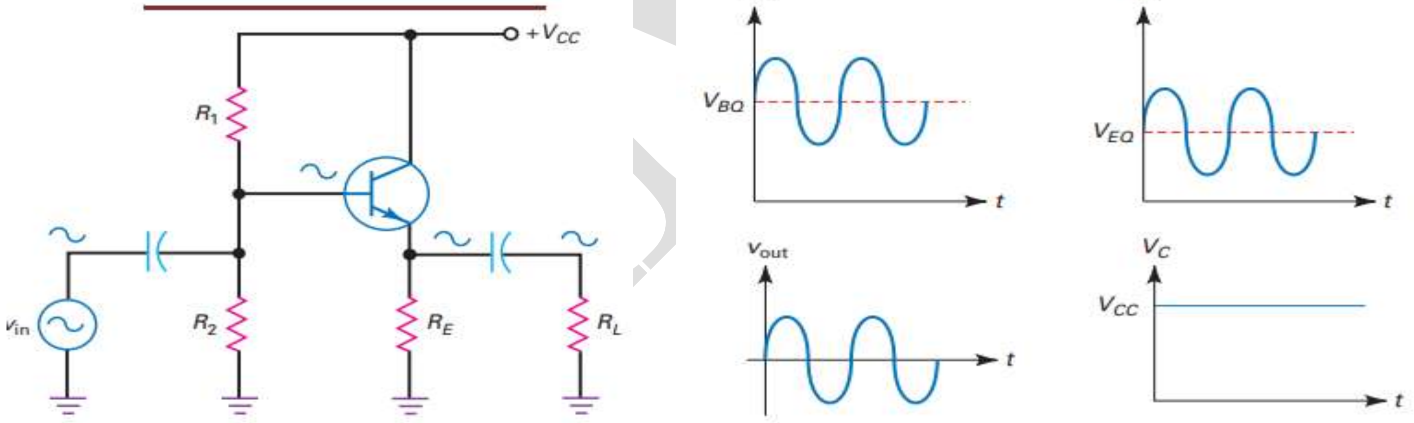
$$\text{Voltage gain } A_V = \frac{v_{out}}{v_{in}} \quad \text{or} \quad v_{out} = A_V v_{in} = 116(1.07) = 124\ mV$$

Common Collector (CC) Amplifier

Q10. With a neat circuit diagram, explain the operation of emitter follower. Also derive the expressions for voltage gain, input and output impedance.

The Emitter follower is also called a common-collector (CC) amplifier. The input signal is coupled to the base, and the output signal is taken from the emitter.

Emitter follower and waveforms.



Operation

Because the collector is at ac ground, the circuit is a CC amplifier. The input voltage is coupled to the base. This sets up an ac emitter current and produces an ac voltage across the emitter resistor. This ac voltage is then coupled to the load resistor. The total voltage between the base and ground has a dc component and an ac component, the ac input voltage rides on the quiescent base voltage V_{BQ}. Similarly the total voltage between the emitter and ground. This time, the ac input voltage is centered on a quiescent emitter voltage V_{EQ}. The ac emitter voltage is coupled to the load resistor.

This output voltage is a pure ac voltage. This output voltage is in phase and is approximately equal to the input voltage. The reason the circuit is called an emitter follower is because the output voltage follows the input voltage.

Since there is no collector resistor, the total voltage between the collector and ground equals the supply voltage. There is no ac signal on the collector because it is an ac ground point.

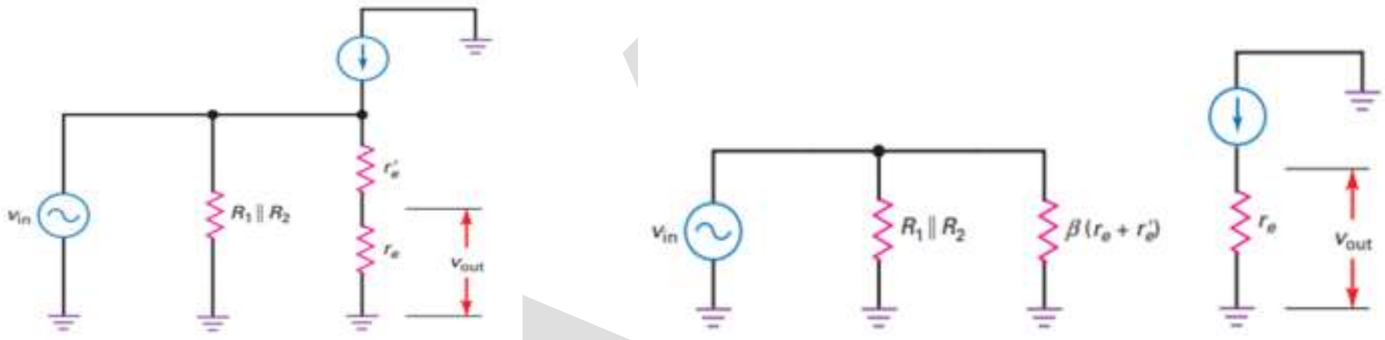
AC Emitter Resistance

The ac signal coming out of the emitter sees R_E in parallel with R_L. The ac emitter resistance is defined as $r_e = R_E || R_L$

Note - This is the external ac emitter resistance, which is different from the internal ac emitter resistance r_e'.

Voltage gain

Figure shows the ac equivalent circuits for a emitter follower using T model and π model.



AC-equivalent circuits for emitter follower.

Using Ohm's law to the B-E circuit

$v_{in} = i_e(r_e + r_e')$ and $v_{out} = i_e r_e$

$$\text{Voltage gain} \quad A_v = \frac{v_{out}}{v_{in}} = \frac{i_e r_e}{i_e(r_e + r_e')} = \frac{r_e}{(r_e + r_e')} \cong 1 \quad \text{for } (r_e \gg r_e')$$

Input Impedance of the Base

Consider the π model

The input impedance of the base : $z_{in(base)} = \beta(r_e + r_e') \cong \beta r_e$

Input Impedance of the Stage

$z_{in(stage)} = R_1 \,||\, R_2 \,||\, \beta(r_e + r_e) \cong R_1 \,||\, R_2 \,||\, \beta r_e$

Output Impedance z_{out} - Consider the T model, connect a test voltage V_X with current I_X at the output terminals so that

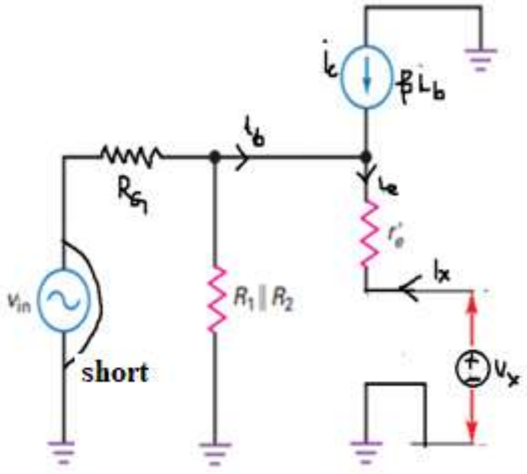
$z_0 = (V_X/I_X)$ and $z_{out} = R_E || z_o$

Applying KVL

$-\,i_b(R_G || R_1 || R_2) - i_e r_e' - V_X = 0$

Negative Feedback

The emitter follower uses negative feedback and the feedback resistance equals all of the emitter resistance and is very high. As a result, the voltage gain is ultrastable, the distortion is almost nonexistent, and the input impedance of the base is very high. Because of these characteristics, the emitter follower is often used as a preamplifier.
The trade-off is the voltage gain, which has a maximum value of 1.



Using $I_X = -i_e$ and $i_e = (\beta+1)i_b \cong \beta i_b$

$$V_X = -i_e\left[\frac{R_G||R_1||R_2}{\beta} + r_e^{|}\right] = I_X\left[\frac{R_G||R_1||R_2}{\beta} + r_e^{|}\right]$$

$$\frac{V_X}{I_X} = z_o = \left[\frac{R_G||R_1||R_2}{\beta} + r_e^{|}\right]$$

$$z_{out} = (R_E||z_o) = R_E||\left[\frac{R_G||R_1||R_2}{\beta} + r_e^{|}\right] \text{--------(A)}$$

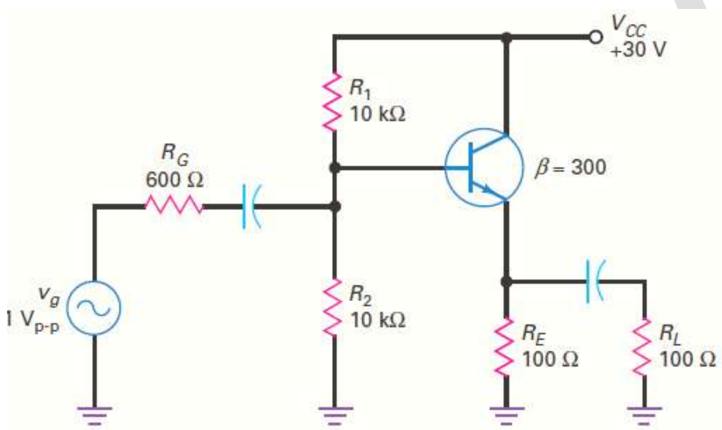
For $(R_G/\beta) << [R_E, (R_1||R_2), r_e^{|}]$

Output Impedance $z_{out} \cong \left[\frac{R_G}{\beta}\right]$

Usually the output impedance is matched to the load so that maximum power is delivered to the load. Choose $z_{out} = R_L$

In this way the effect of R_L on the output voltage is minimized and the circuit is acting as a buffer between the input and output (Impedance matching).

Problem – Determine the output impedance of the circuit shown below



Solution

$R_B = R_{Th} = R_1||R_2 = 10||10 = 5 \text{ k}\Omega$

$$V_B = V_{Th} = \frac{V_{CC}R_2}{R_1+R_2} = \frac{30(10)}{10+10} = 15 \text{ V}$$

Assuming $V_{BE} = 0.7$V and given β=100

$$I_{E1} = \frac{V_B - V_{BE}}{R_E + \frac{R_B}{\beta+1}} = \frac{15-0.7}{0.1+\frac{5}{301}} = 138 \text{ mA}$$

$$r_{e1}^{|} = \frac{25 \text{ mV}}{I_E} = \frac{25 \text{ mV}}{138\times10^{-3}} = 0.18\,\Omega$$

$z_{in(base)} = (R_G||R_1||R_2) = [0.6||10||10] = 536\Omega$

$$z_{out} = R_E||\left[\frac{R_G||R_1||R_2}{\beta} + r_e^{|}\right] = 100\,||\,\left[\frac{536}{300} + 0.18\right] = 1.9\,\Omega$$

Verification Output Impedance $z_{out} \cong \left[\frac{R_G}{\beta}\right] = \left[\frac{600}{300}\right] = 2\,\Omega$

Review Questions

1. Explain how Coupling Capacitors and Bypass Capacitors are chosen for Electronic amplifier circuits.
2. Explain the operation of (i) Base Biased amplifier (ii) VDB amplifier.
3. What is emitter resistance and how is determined?
4. Briefly explain the transistor models.
5. Draw the ac equivalent circuit for (i) Base Biased amplifier (ii) VDB amplifier.
6. With a neat circuit diagram and relevant equivalent circuits, explain CE amplifier and obtain the expression for voltage gain.
7. Explain the working of CC amplifier and derive the expression for (i) Voltage gain (ii) input and (iii) output impedances